Pylint before:

```
SE_Lab_PES2UG24CS456 > lab5 > pylint_before.txt
1   ************* Module music_catalog
2   music_catalog.py:10:0: C0301: Line too long (109/100) (line-too-long)
3   music_catalog.py:1:0: C0114: Missing module docstring (missing-module-docstring)
4   music_catalog.py:9:0: C0116: Missing function or method docstring (missing-function-docstring)
5   music_catalog.py:9:0: C0103: Function name "addSong" doesn't conform to snake_case naming style (invalid-name)
6   music_catalog.py:9:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
7   music_catalog.py:10:17: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
8   music_catalog.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
9   music_catalog.py:15:0: C0103: Function name "shufflePick" doesn't conform to snake_case naming style (invalid-name)
10  music_catalog.py:20:0: C0116: Missing function or method docstring (missing-function-docstring)
11  music_catalog.py:22:4: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
12  music_catalog.py:28:0: C0116: Missing function or method docstring (missing-function-docstring)
13  music_catalog.py:28:0: C0103: Function name "savePlaylist" doesn't conform to snake_case naming style (invalid-name)
14  music_catalog.py:32:0: C0116: Missing function or method docstring (missing-function-docstring)
15  music_catalog.py:32:0: C0103: Function name "loadPlaylist" doesn't conform to snake_case naming style (invalid-name)
16  music_catalog.py:36:0: C0116: Missing function or method docstring (missing-function-docstring)
17  music_catalog.py:36:0: C0103: Function name "totalPlays" doesn't conform to snake_case naming style (invalid-name)
18  music_catalog.py:39:4: W0702: No exception type(s) specified (bare-except)
19  music_catalog.py:43:0: C0116: Missing function or method docstring (missing-function-docstring)
20  music_catalog.py:48:19: C0321: More than one statement on a single line (multiple-statements)
21
22  -----------------------------------
23  Your code has been rated at 4.57/10
24
25
```

Bandit before:

```
SE_Lab_PES2UG24CS456 > lab5 > bandit_before.txt
1   [main]   INFO  profile include tests: None
2   [main]   INFO  profile exclude tests: None
3   [main]   INFO  cli include tests: None
4   [main]   INFO  cli exclude tests: None
5   [main]   INFO  running on Python 3.13.7
6   Run started:2026-10-14 06:35:52.243565+00:00
7
8   Test results:
9   >> Issue: [B403:blacklist] Consider possible security implications associated with pickle module.
10     Severity: Low   Confidence: High
11     CWE: CWE-502 (https://cwe.mitre.org/data/definitions/502.html)
12     More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_imports.html#b403-import-pickle
13     Location: .\music_catalog.py:1:0
14  1  import pickle
15  2  import random
16  3  import sqlite3
17
18  --------------------------------------------------
19  >> Issue: [B608:hardcoded_sql_expressions] Possible SQL injection vector through string-based query construction.
20     Severity: Medium   Confidence: Medium
21     CWE: CWE-89 (https://cwe.mitre.org/data/definitions/89.html)
22     More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b608_hardcoded_sql_expressions.html
23     Location: .\music_catalog.py:10:17
24  9  def addSong(title, artist, plays=0, playlist=[]):
25  10     conn.execute("INSERT INTO songs (title, artist, plays) VALUES ('%s', '%s', %d)" % (title, artist, plays))
26  11     playlist.append(title)
27
28  --------------------------------------------------
29  >> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
30     Severity: Low   Confidence: High
31     CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
32     More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
33     Location: .\music_catalog.py:17:11
34  16     ids = [r[0] for r in conn.execute("SELECT id FROM songs")]
35  17     return random.choice(ids)
36  18
37
38  --------------------------------------------------
39  >> Issue: [B301:blacklist] Pickle and modules that wrap it can be unsafe when used to deserialize untrusted data, possible security issue.
40     Severity: Medium   Confidence: High
41     CWE: CWE-502 (https://cwe.mitre.org/data/definitions/502.html)
42     More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b301-pickle
43     Location: .\music_catalog.py:33:11
44  32 def loadPlaylist(blob):
45  33     return pickle.loads(blob)
46  34
47
48  --------------------------------------------------
49
50  Code scanned:
51     Total lines of code: 35
52     Total lines skipped (#nosec): 0
53     Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0
54
55  Run metrics:
56     Total issues (by severity):
57        Undefined: 0
58        Low: 2
59        Medium: 2
60        High: 0
61     Total issues (by confidence):
62        Undefined: 0
63        Low: 0
64        Medium: 1
65        High: 3
66  Files skipped (0):
67
```

Flake8 before:

SE_Lab_PES2UG24CS456 > lab5 > flake8_before.txt

```
music_catalog.py:6:80: E501 line too long (99 > 79 characters)
music_catalog.py:10:80: E501 line too long (109 > 79 characters)
music_catalog.py:21:80: E501 line too long (86 > 79 characters)
music_catalog.py:39:5: E722 do not use bare 'except'
music_catalog.py:47:5: E741 ambiguous variable name 'l'
music_catalog.py:48:18: E701 multiple statements on one line (colon)
music_catalog.py:49:80: E501 line too long (83 > 79 characters)

```

Pylint after:

SE_Lab_PES2UG24CS456 > lab5 > pylint_after.txt

```

---
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)


```

Bandit after:

SE_Lab_PES2UG24CS456 > lab5 > bandit_after.txt

```
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.7
Run started:2026-10-14 06:45:11.634201+00:00

Test results:
    No issues identified.

Code scanned:
    Total lines of code: 63
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
Files skipped (0):

```

Flake8 after:

SE_Lab_PES2UG24CS456 > lab5 > flake8_after.txt

```
Generate code (Ctrl+I), or select a Language (Ctrl+K M). Start typing
```